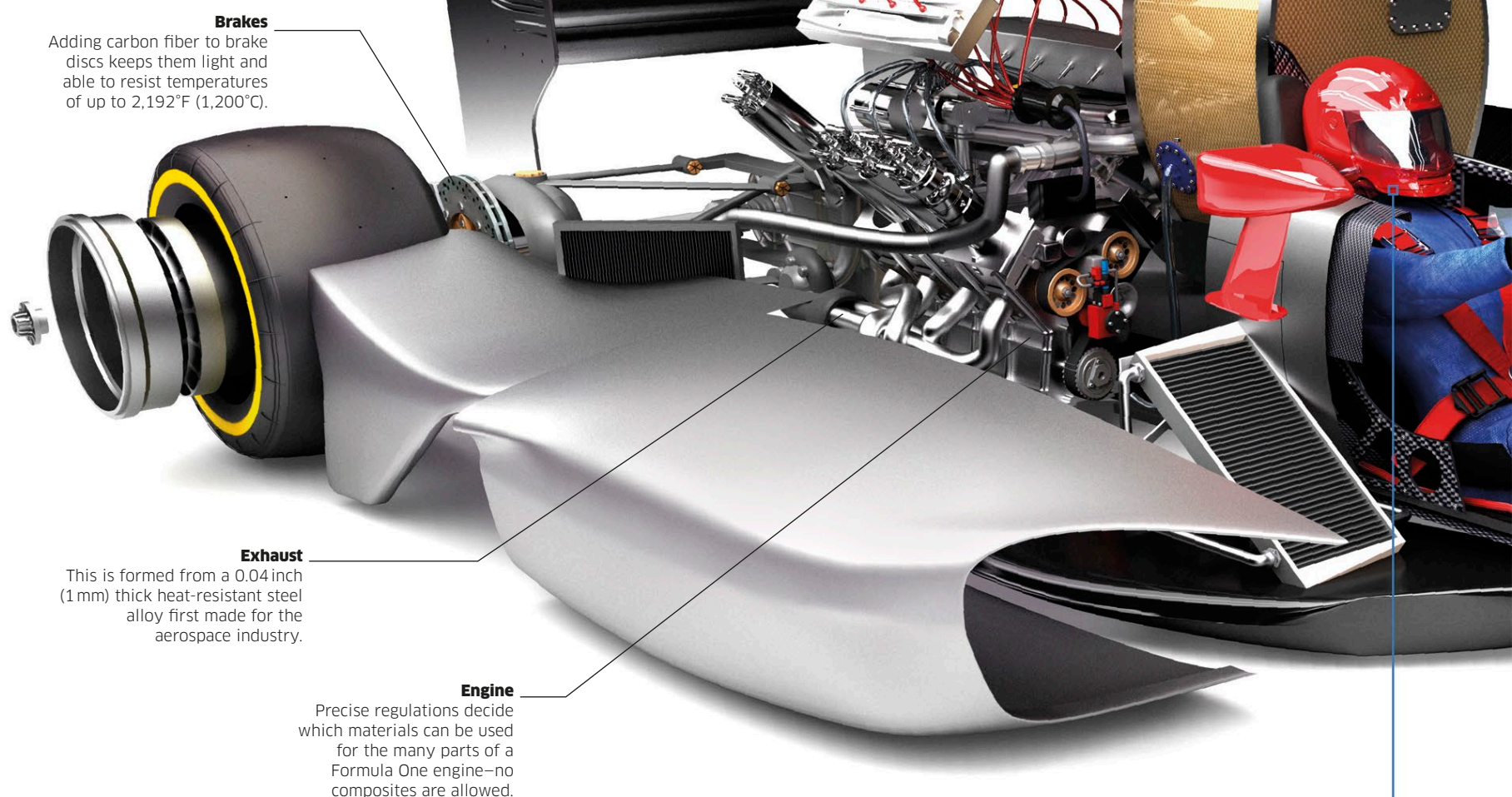


Materials technology

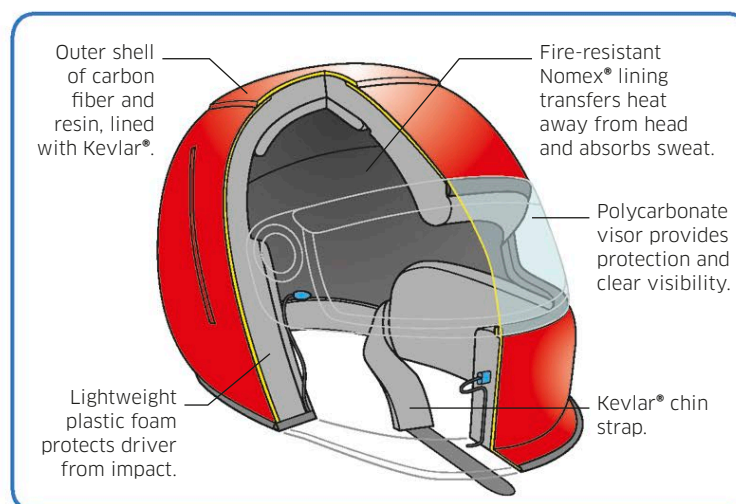
Synthetic materials are born in laboratories. Using their knowledge of elements and compounds, chemists can develop new materials with unique properties, created for specific tasks.

Materials created artificially perform different functions depending on their chemistry—the arrangements of their atoms or molecules, and how they react. Research constantly brings new materials to meet new challenges, ranging from synthetic textiles and biodegradable plastics to the vast range of high-performance materials that make up a racing car.



Racing car

Formula One cars rely on materials that can withstand extreme heat and pressure. The structure must be rigid in some parts and flexible in others; some parts are heavy while some have to be light. The drivers are also exposed to heat and pressure—and speeds over 200 mph (320 km/h)—and rely on synthetic materials to keep safe. Their clothing is made with layers of Nomex®, a fire-resistant polyamide (a type of plastic) used for fire and space suits. Kevlar®, similar to Nomex® but so strong it is bullet proof, is used to reinforce various car parts as well as the driver's helmet.



Helmet anatomy

Drivers are subjected to extreme G-forces when braking and cornering. This puts great strain on their necks. To help keep their heads up, their helmets must be as light as possible. Highly specialized materials are used for the helmets, which need to be light and comfortable, yet strong and able to absorb impacts and resist penetration in case of an accident.